

Vermont Mesh Project

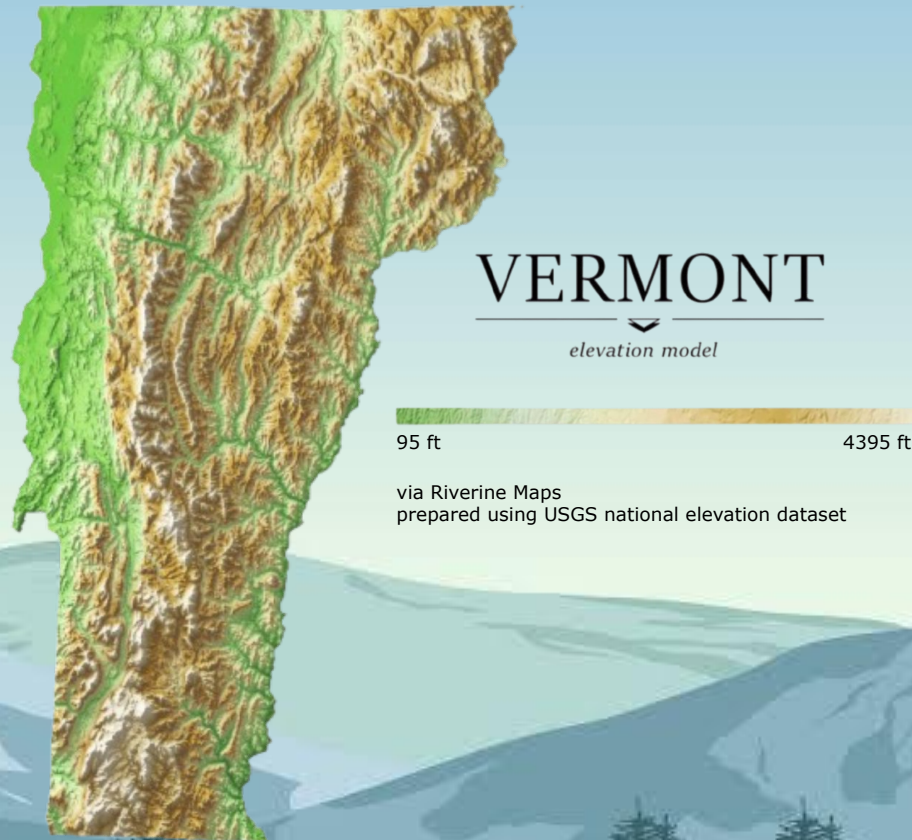
Gavin Akersten, Gabe Christiansen



University
of Vermont

What are we doing?

Our mission: build an open-source, long-range (LoRa) radio network that keeps Vermont residents connected when cell towers and power grids fail.



What problem exists?

Vermont's mountainous terrain already strains communication infrastructure — and when it fails during a natural disaster, residents are left completely cut off. No ability to call for help or communicate with their neighbors or family members. In a serious event, people could go days without word from each other, or the ability to attain essential resources.



2023: floods in the Vermont state capital via CNN



2014: Damage from ice storm blocks major roads

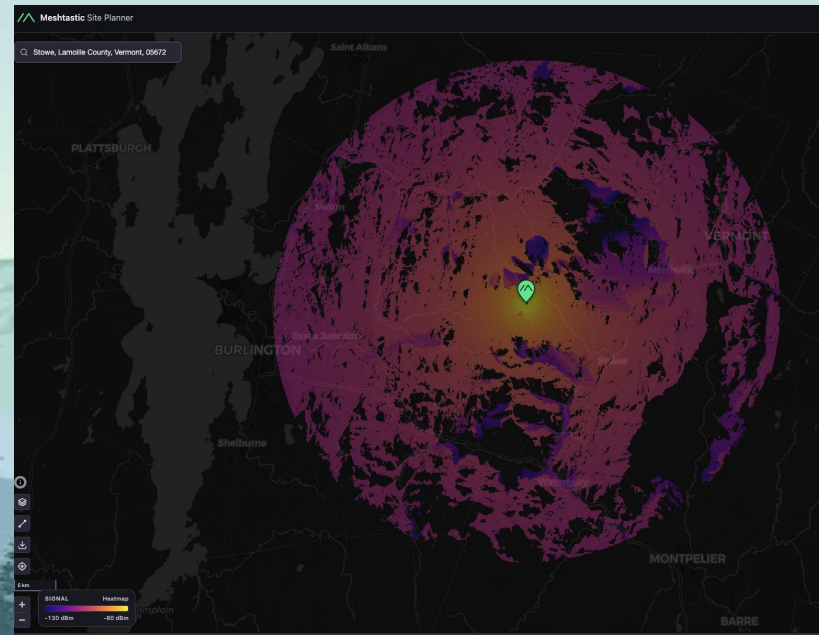
What is the solution?

Set up a network of nodes at points of high elevation and good sight lines across the state to enable off grid communication at times of distress.

Starting point: UVM Votey Hall

Theoretical node locations:

- Killington Peak Lodge
- Jay Peak Aerial Tram
- Mount Mansfield Peak Visitors Center
- Mount Equinox Radio Towers
- Burke Mountain fire towers



LoRa node coverage simulated from the peak of Mount Mansfield

Why is this important?

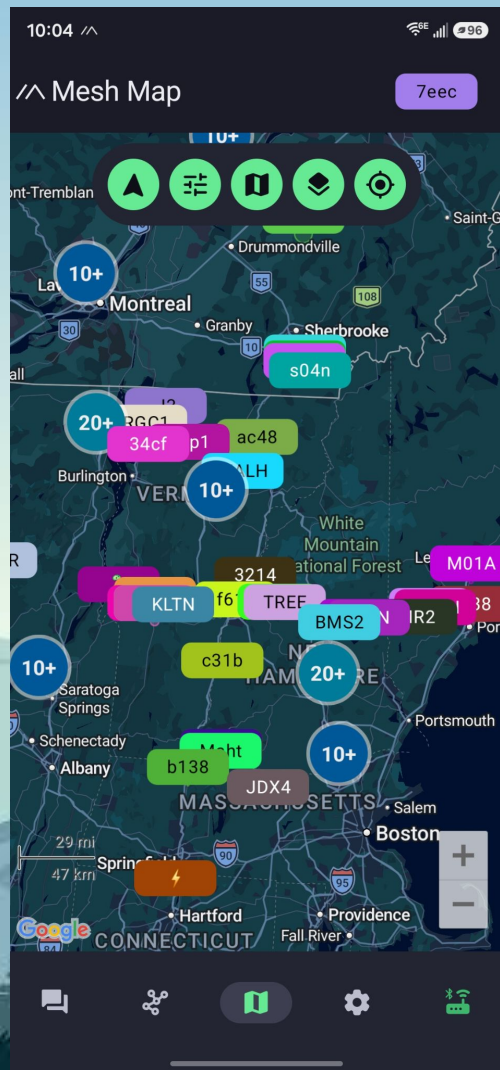
Natural disasters are frequent, unpredictable, and carry real human cost. Our network gives people a safety net to fall back on when the power grid goes down, and it costs a fraction of traditional emergency infrastructure. The time to build it is now: every disaster we wait through costs more lives and more resources.



Why will we succeed?

We're not starting from scratch. Meshtastic already has an active community of developers, researchers, and users driving innovation. And in Vermont, the incentives align: partners get an off-grid communication system for the state, we get access to install nodes.

As students and residents of this state, we have witnessed the effects of harsh weather. We are passionate about making our state safer and more resilient



What are we asking for?

We're asking for funding to purchase Meshtastic nodes at roughly \$50–\$100 each. Phase 1 needs 3–5 nodes to pilot testing in Burlington; scaling statewide would run approximately \$5,000.

We're also seeking partners: ski patrols, who can give us access to peak lodges in exchange for using the devices for emergency rescue, and cities and state parks, who stand to benefit directly from off-grid communication while supporting the network's reach.

